

A Novel Energy Management Algorithm for Solar Based Electric Vehicle

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Abstract—A solar based electric vehicle (EV) requires optimal use of the vehicle roof-top space. A distributed PV array would be good solution for the solar based EV system. In this paper, performance of different rated photovoltaic (PV) array with individual DC-DC converter and MPPT algorithm which are connected to the common DC bus are investigated. The battery charging profile from both solar and grid are investigated. To optimal use of the PV power and to operate the battery in safe region a novel energy management algorithm is proposed. The motor of vehicle will energized from both the PV and battery system. In this algorithm, both grid to vehicle (G2V) and vehicle to grid (V2G) features are incorporated. The complete simulation study is carried out in MATLAB/SIMULINK platform

Keywords—Distributed MPPT, electric vehicle, energy management, photovoltaic

I. INTRODUCTION

Nowadays, the solar energy is the most promising power source for replacement of conventional energy sources such as fossil fuel, gas, coal etc. In world, the transport sector consumes much power which causes environmental pollution. To minimize the pollution, alternative fuel source based vehicles are developed. These vehicles are categorized as pure electric vehicle (EV), hybrid electric vehicle (HEV), and plug-in hybrid electric vehicle (PHEV). Battery system is an integral part of all the types of EV. In high power EVs, the battery modules are connected in series/parallel to meet the power demand. To operate at its maximum efficiency and also to increase its life cycle, a good charging system is required. In a conventional charging systems, AC source is used as input. But during peak hour, it causes an extra burden to the grid. The battery can also be charged through the renewable energy sources such as solar or wind or fuel cell. But, the solar and fuel cell can be used for dynamic charging of the vehicle. Among the renewable power sources, solar power is abundantly available and also installation is easy.

The PV system can be used to charge the battery by mounting it on roof of the vehicle (on board charging), or a solar based charging station. In [1], a brief design of solar based charging station is discussed. Based on the rating of the charging station, suitable PV array are taken. In an on-board solar based charging system, the PV modules are mounted on the roof of the vehicle for charging the battery or some specific application of vehicles. Due to this, the driving

ranges of the EVs can be extended. A detailed design structure of a solar assisted three-wheeler is presented in [2]. However, the design is limited to a standalone system with a single PV module on the roof of the vehicle. In [3], a detailed study on impact on the output power by installing the PV cells on the roof of a vehicle are given. Different types of charging topologies and methods are highlighted in [4-5]. As per author's knowledge, very few research works have been undertaken on the solar based on-board EV system. In the on-board system, the PV power generation capacity is generally low in capacity so it is not sufficient to meet completely charge the battery. It is basically used for extending the travel distance range.

The power generation of solar energy is very intermittent in nature. So a maximum power point tracking (MPPT) algorithm is required to extract maximum power from the PV system. In most of the PV system, perturb and observe (P&O) or incremental conductance (INC) are used. The P&O algorithm has inherent problem of oscillating around the MPP. It may not be that much suitable for solar based EV system. In many of the existing literatures, the PV system is used to charge the battery but the MPPT condition is not much highlighted. However, in a PV mounted vehicle, the vehicle speed depends on the road profile. Similarly, the PV power generation depends on the solar insolation. The dynamic charging or discharging condition has not been investigated so far. In an EV, motor plays a very vital role. The solar vehicle is basically for low to medium ranges of application. A brief review of different motors for EV application are discussed in [6]. In an EV, battery is the primary source of storage system. To optimize the use of PV and smooth running of motor, some auxiliary power sources such as super-capacitor or fuel cell are added with the battery. However it increases system cost. In [7], PV, battery, and super-capacitor based EV system is presented and also an energy management algorithm for optimal use of the power sources are discussed. However, this system is also for standalone purpose. In most of the reported literature, the battery of the EV charges from the AC sources. The renewable sources application for EV with dynamic charging system are given in [8-9]. In some cases to optimize the use of battery, during rest time of vehicle the battery power transfers to grid (V2G). More detailed review and design approach of V2G and vehicle to vehicle and vehicle to home are discussed in [10].

In general, the roof of the vehicle is not uniform in shape. In this paper, a different rating of the PV array for different places of the roof of the vehicle are considered. All the PV arrays with MPPT are connected in series to form a PV system. The PV system is integrated to the common DC-link capacitor. This configuration is called distributed MPPT (DMPPT) type of PV system. In this study, the performance of DMPPT based PV system for battery charging and for motor drive is investigated. A universal charging system is developed by considering different operating scenarios of the system. A novel energy management algorithm with control structure is proposed for DC voltage regulation and also protection of battery. The performance of the vehicle are evaluated under five different conditions as follows:

1. Solar radiation available and battery is fully charged state, vehicle is in moving conditions
2. Solar radiation available, battery is in fully charged state, vehicle is in standing on the way
3. Power transfer from PV to grid
4. Battery Charging from grid support
5. Transferring battery power to the grid

The complete study is described as follows. The complete system is described in Section II. Control structure of each sources and the proposed energy management algorithm are explained in section III. Results are discussed in Section IV and conclusion of this paper is given in Section V.

II. SYSTEM DESCRIPTION

The complete system of the study is shown in Fig. 1. In a car (four wheeler), the solar module may be mounted on different place of the body of the car. In this study, front, middle and back sides on the roof of the vehicle are considered. Due to the different area of each position, different rated PV array can be installed. In this system, the series connected PV array with individual boost converters are integrated to the common DC-link capacitor. To increase settling time of the MPPT and also to overcome the drawbacks of P&O MPPT, a Fibonacci search based MPPT algorithm is used to extract the maximum power from each PV array [11].

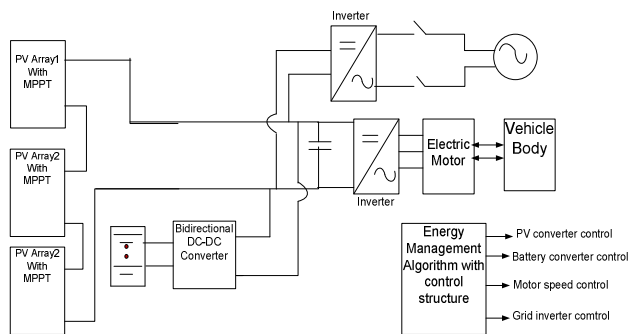


Fig. 1: Complete system of solar based electric vehicle

Battery of the vehicle is connected to the common DC link capacitor through a bidirectional DC-DC converter. An induction motor based vehicle is considered. It is connected to the DC-link capacitor through a voltage source inverter. To charge the battery during night or cloud condition, there is provision of charging through the grid. The complete circuit is designed to integrate it to the grid system for charging and also power transfer from the PV system to the grid during day time. The battery can also transfer the power to the grid during rest state of the vehicle whenever required.

All the possible approaches of power management are considered in this article.

III. CONTROL STRUCTURES IN THE SYSTEM

In a solar mounted vehicle, power can be drawn from both the PV and battery system to drive the motor. In this system, the battery continuously charge and discharge based on the SoC condition and the speed of the vehicle. The objective of the control structures are to maintain the desired voltage across the DC-link capacitor, speed control of induction motor, to operate the battery in safe region of charging and discharging, extracting maximum power from the PV array, and transferring the PV generated power and battery power (when required) without disturbing power quality at the grid side. During vehicle movement, the DC-link capacitor voltage is regulated by using the DC-DC bidirectional converter of the battery system. When the vehicle is rest mode in the allocated place, the PV system is used to charge the battery and remaining extra power will fed to the grid. To optimize the power from each sources a suitable energy management algorithm is required. The complete operation of the system is categorized into four different modes. The energy management algorithm for mode-1 and 2 is shown in Fig. 2.

Mode 1: Solar radiation available and battery is in fully charged state, vehicle is in moving conditions:

In this mode, considered as the battery is in fully charged state, so maximum available PV power is used to drive the motor. The vehicle speed depends on the drive cycle of the city. Thus, the power demand by the vehicle varies accordingly. As the PV panels are mounted according to the available space on the roof of the vehicle and each panel has individual power converter for MPPT, so the power generation from each section of the PV array is also different. In this proposed energy management algorithm, when the surplus PV power is generated and SoC of battery is more than 90%, initially the low rated PV array will operate at open circuit voltage. Again the battery SoC is checked if it is increasing continuously, next low rated PV array will operate in open circuit voltage. This condition will continue till the SoC of battery remains constant or in the safe range of upper limit. The operation of the PV array of the PV system is controlled by the energy management algorithm as shown in Fig. 2.

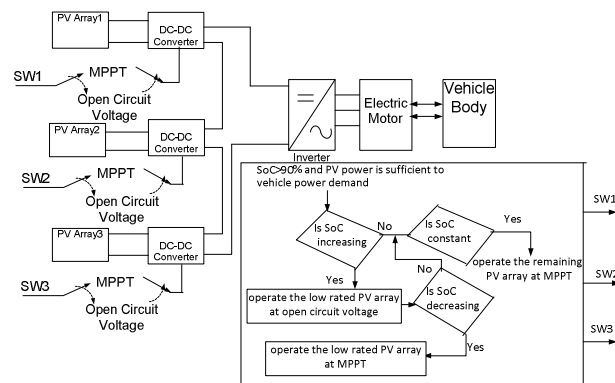


Fig. 2: Energy management algorithm for Mode-1 and 2

Mode 2: The vehicle is standing mode on the way:

The PV system generates power based on the solar radiation. When the vehicle is standing/rest mode and if the battery is in fully charged, complete PV array will operate at it operates

open circuit voltage. If the battery power is below the upper limit of the SoC, based on the level of the SoC the PV system will operate at its maximum power.

Mode 3: The vehicle is on parking place during day time:

During day time, the PV system can generate its maximum power. As the vehicle is on parking place so no power is used by the motor. During this condition only the battery absorbs the PV power. When the battery becomes fully charged at this condition the PV power will be unused. To optimize the use of PV power, the surplus PV power should be transferred to the grid. In this study, an on-board circuit with inverter and LC filter are integrated in the vehicle. The advantages of such as configuration as the vehicle can be charged residential place, no such specific parking station is required. The PV system power depends on solar insolation. When the vehicle is on standstill and battery is on fully charged state, the PV generated power is not used. During that condition, the vehicle may be plugged to the grid to transfer the surplus PV generated power to the grid. To operate the battery in safe mode a current control approach is used to charge the battery. In this controller, battery can be controlled at its rated current capacity. The control structure of the bidirectional DC-DC converter is controlled to operate the battery in safe region.

Mode 4: The battery is in discharged mode during night time/cloud day:

During cloud day, the battery may be in discharged state. In general, the vehicle is parked at the allocated parking slot. In this study, it is considered that the grid plug is available at the parking slot. During night time at parking place, the vehicle battery should be charged through the grid. A suitable control structure is developed to charge the battery through grid support. The DC-link capacitor voltage is maintained through grid support. This feature is also called as G2V.

Mode 5: Vehicle to Grid condition:

During the peak hour of the day basically at evening time, more power are drawn from the grid. If the vehicle battery is in fully charged and in steady state, the vehicle may transfer stored energy to the grid.

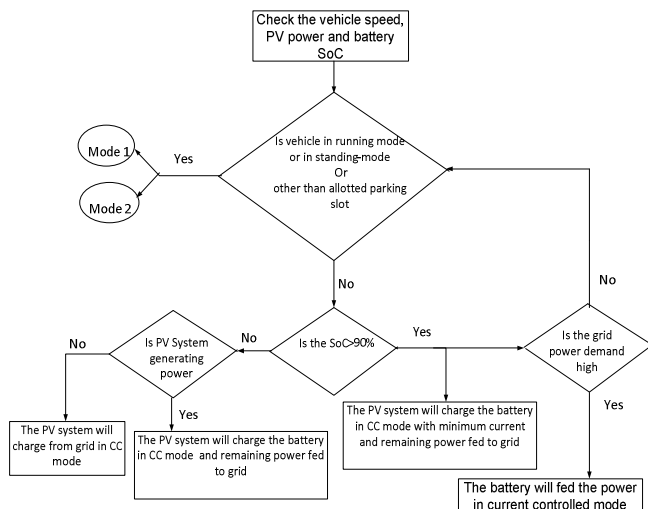


Fig. 3: Energy management algorithm for Mode 3 and 4

This feature is called as vehicle to grid. The DC-link capacitor voltage also maintained by the grid support. Based on the rating of battery, a current control approach is used to transfer the power from the battery to grid. In general, the V2G feature is used in high power EV system. To verify the concept the same feature is given in this study.

The control structure of complete system divided into following blocks: MPPT of system, voltage and current control for bidirectional DC-DC converter to operate the battery in the safe region. A variable voltage and variable frequency (VVVF) method is used for speed control of induction motor, and current control technique is used for the PV power transfer to the grid and also DC link voltage regulation without compromising the power quality.

The closed loop v/f control approach is used to control the speed of the induction motor. The switching pulses for the inverter is generated using PWM techniques. The switching frequency of the IGBT's depends on the speed error which i_d is the differences between reference and actual speed. In this control approach, the rotor speed is measured and same is compared to the reference speed. In general, the reference speed is the drive cycle. The difference between reference and actual speed is considered as error signal. It is passed through a proportional-integral (PI) controller. In this study, the controller parameters are tuned in trial and error method.

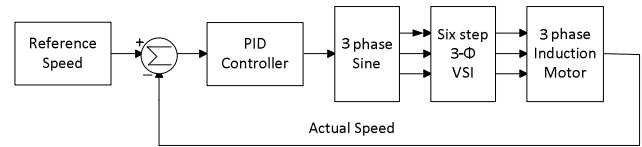


Fig. 4: Block diagram for speed control of motor of vehicle [12]

Block of the speed controller is shown in Fig. 4. More detailed of this controlled are given in [12]

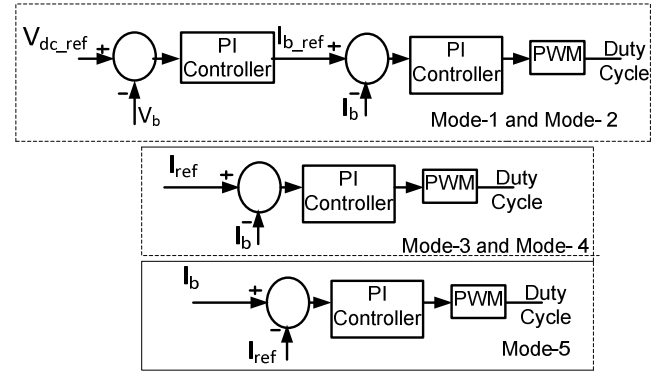


Fig. 5: Bidirectional DC-DC converter control

Under vehicle parking on allocated place at day time the PV system will fed the power to the grid. In this study, double loop control approach is used for DC-link voltage regulation and tracking of desired current at inverter output and also for maintain the power quality at grid side. The control structure of bidirectional DC-DC under all operating mode is given in Fig. 5. In mode 1 and 2, it is controlled as voltage control approach. In other modes current control approach is used. More detailed explanation of the control structure of a bidirectional DC/DC converter is given in [13]. In this study, a conventional double loop control structure such as outer loop is for DC link voltage regulation and inner loop is for

inverter current control. More detailed description of inverter control is given in [14].

IV. SIMULATION RESULTS AND DISCUSSION:

The complete system of study is shown in Fig.1 In this study, the rating of different parameters are shown in Table 1. In this simulation study, a 96V, 100Ah battery is taken. However in standard 24V or 48V with higher current rating can also be used.

Table 1: System parameters of Simulation study

Components	Rating
PV array	520W, 440W, and 350W
Battery	96V, 100Ah
DC-link reference voltage	350V
Induction Motor	3HP
Grid Voltage	230V

The vehicle is connected to grid through the electric plug of a specified parking slot. During running condition of the vehicle, both battery and PV system will energize the motor. The complete power managements are classified in five different modes. The simulation study is carried out by using MATLAB/SIMULINK. The simulation results of each mode is discussed below.

Mode 1 and 2:

In this mode, the vehicle is on running state, the solar and battery system supplies power to the motor of the vehicle. The voltage across the DC-link capacitor is shown in Fig. 6(a). It is regulated by using a double loop control approach [12]. Based on the speed and SoC of battery the PV arrays are selected. The DC-link voltage is also regulated under transition of the PV array. The PV array power is shown in Fig. 6(b). In this system, three different rated PV arrays are used. Based on the availability of the spaces on the roof of the vehicle the PV panels will be mounted. In this simulation study, 520W, 440W, and 350W rating of PV array are considered. Initially, all the PV arrays are operated at their MPPT voltage. As the PV system generated power is sufficient to drive the motor of the vehicle and also the SoC of battery reaches to upper charging limit, at this point the low rated PV array operates at its open circuit voltage. As a result, the charging rate of the battery decreases as shown in Fig. 6(c). As per the figure, other PV arrays are also forced to operate at its open circuit voltage because the vehicle demands only very less power. Again when vehicle demands more power due to increase of speed, the battery SoC starts decreasing. When the battery SoC falls below a certain limit, again the low rated PV array is operated at its MPP. This process continues till the battery operates in the safe region. In this study, for demonstration purpose the upper and lower limit of SoC are taken as 90.025 and 90.01%, respectively. The reference and actual speed of the vehicle are shown in Fig. 6(c). The reference speed and actual speed of the motor is shown in Fig. 6(d). The V/f control approach completely track the reference speed.

When the vehicle is on static state on the road, very minimum power is required for ancillary services of the vehicle. At this scenario, the SoC of the battery is maintained in safe region by following the same approach as mentioned in Fig. 6 (a).

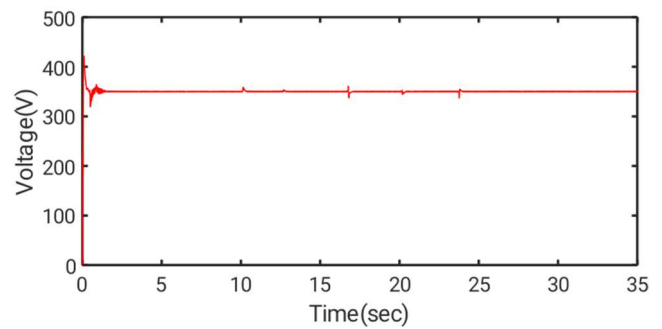


Fig. 6(a): Voltage across the DC-link capacitor

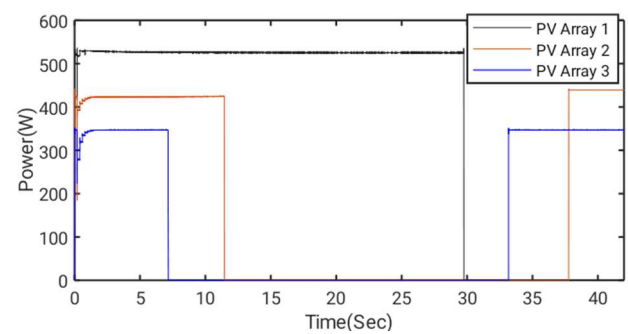


Fig. 6(b): PV system Power

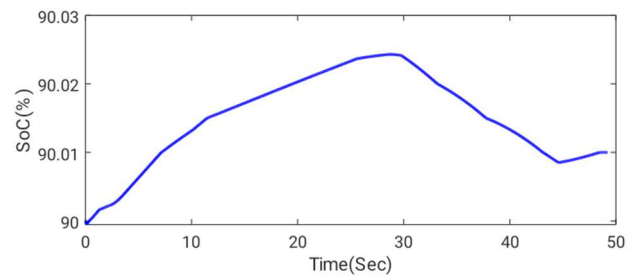


Fig. 6(c): SoC of Battery

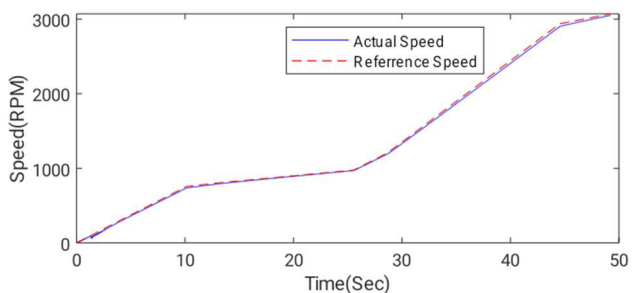


Fig 6(d): Actual and reference speed of the vehicle

Mode 3 and 4:

In general, during night time or some duration of the day the vehicles are parked in the allocated parking slot. To optimize the PV generated power, an on-board inverter control with grid-tied system are available in the vehicle. During rest time of the vehicle it is plugged to the grid. In day time, the PV system generates power uses to charge the battery and remaining power transfers to the grid. In this study, based on the SoC of the battery, a current control structure is used to charge the battery. It is assumed that the battery is not fully charged state. So initially, it will be charged at its rated current and slowly the reference current will decrease. The reference and actual charging of the battery are shown in Fig. 7(a). Both the currents are nearly same as shown in figure. The PV generation power is considered as constant. When the

PV system power is not sufficient to charge the battery, the grid provides the surplus power to charge the battery at a desired current rate as shown in Fig. 7(b). So the battery can be charged both from the grid and PV system. The surplus generated power transfers to the grid. The SoC of battery is shown in Fig. 7(c). When SoC of the battery reaches at a particular limit, the charging current is taken as minimum value. As the power fed to the grid and also drawn from the grid for charging purpose, the power quality should not be distorted. The reactive power at the grid side is shown in Fig. 7(d). It indicates that the grid voltage and current are in phase. Voltage across the DC-link capacitor is shown in Fig. 7(e). This mode is also referred as grid to vehicle (G2V) mode.

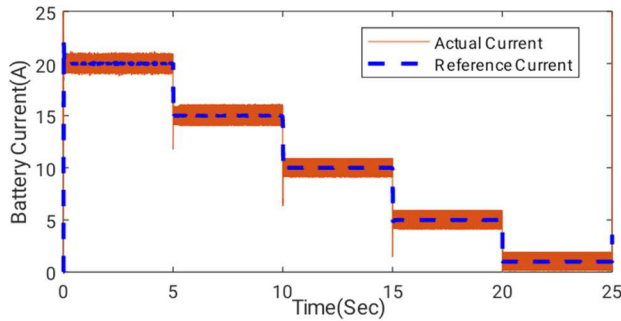


Fig. 7(a): reference and actual charging current

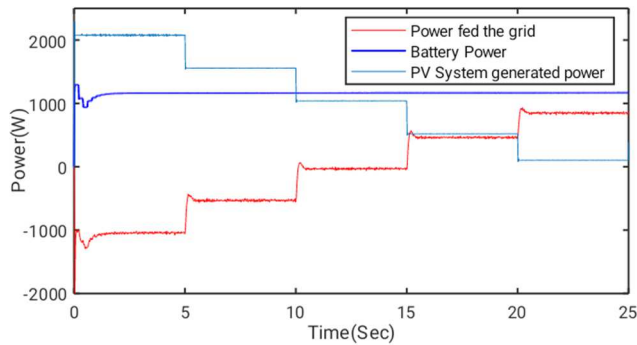


Fig. 7(b): PV System, Battery, and grid power

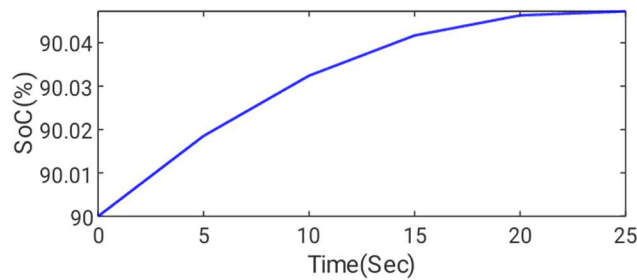


Fig. 7(c): SoC of battery

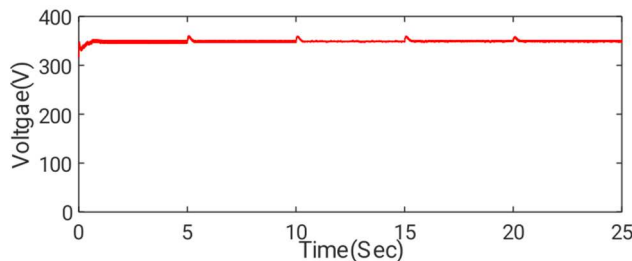


Fig. 7(d): Voltage across DC-link capacitor

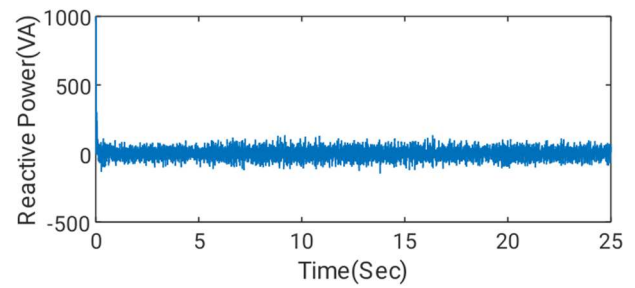


Fig. 7(e): Reactive Power at the grid side

Mode 5:

In this mode, the battery power of the vehicle will transfer to the grid. It is also called as vehicle to grid (V2G) mode of operation. In this study, different rating of current is set to transfer the power from the battery to grid. Initially, more current is set and the reference current decreases with respect to time. Accordingly, the power fed to the grid also varies. The battery system power, PV power, and power transferred to the grid are shown in Fig. 8(a). The PV system power is considered as constant, so power transfer to the grid varies with battery system generated power. The SoC of the battery is shown in Fig. 8(b)

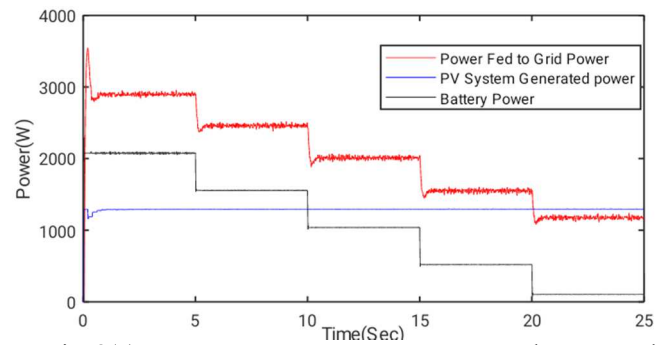


Fig. 8(a): Battery power, PV system generated power and power fed to the grid

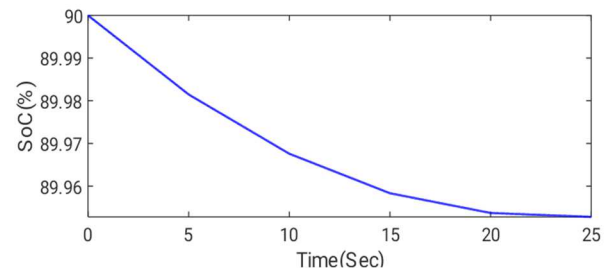


Fig. 8(b): SoC of battery

V. CONCLUSION

This study presents a novel energy management algorithm of a solar mounted on-board charged electric vehicle. The performance of different solar array with individual converter with MPPT algorithm i.e. distributed MPPT are found satisfactory for electric vehicle application. The distributed MPPT based PV array found satisfactory for charging the battery under rest mode of the vehicle and also to drive the motor during moving condition of the vehicle. The operation of complete PV system are used in optimal way to operate the battery in the safe region and also to drive the motor of the vehicle.

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